



UNITS AND SYMBOLS IN SOLAR ENERGY

In 1977, a committee of ISES developed a set of recommended nomenclature for papers appearing in *Solar Energy*. This is a condensed and revised version of those recommendations. The original appeared in *Solar Energy* **21**, 61–68 (1978). Copies of complete recommendations are available from the editorial offices.

1. UNITS

The use of S.I. (Système International d'unités) in *Solar Energy* papers is mandatory. The following is a discussion of the various S.I. units relevant to solar energy applications.

Energy

The S.I. unit is the joule ($J \equiv \text{kg m}^2 \text{s}^{-2}$). The calorie and derivatives, such as the langley (cal cm^{-2}), are not acceptable. No distinction is made between different forms of energy in the S.I. system so that mechanical, electrical, and heat energy are all measured in joules. Although the watt-hour is used in many countries for commercial metering of electrical energy, its use is discouraged in scientific and technical papers as it is derived from the hour which is not a basic S.I. quantity.

Power

The S.I. unit is the watt ($W \equiv \text{kg m}^2 \text{s}^{-3} \equiv \text{J s}^{-1}$). The watt will be used to measure power or energy rate for all forms of energy and should be used wherever instantaneous values of energy flow rate are involved. Thus, energy flux density will be expressed as W m^{-2} and heat transfer coefficient as $\text{W m}^{-2} \text{K}^{-1}$. Energy rate should not be expressed as J h^{-1} .

When power is integrated for a time period, the result is energy that should be expressed in joules, e.g. an energy rate of 1.2 kW would produce $1.2 \times 3600 = 4.3 \text{ MJ}$ if maintained for 1 h. It is preferable to say that

$$\text{Hourly energy} = 4.3 \text{ MJ}$$

rather than

$$\text{Energy} = 4.3 \text{ MJ h}^{-1}.$$

Force

The S.I. unit is the Newton ($N \equiv \text{kg m s}^{-2}$). The kilogram weight is not acceptable.

Pressure

The S.I. unit is the Pascal ($\text{Pa} \equiv \text{N m}^{-2} \equiv \text{kg m}^{-1} \text{s}^{-2}$). The unit kg cm^{-2} should not be used. It is sometimes practical to use $10^5 \text{ Pa} = 1 \text{ bar} = 0.1 \text{ MPa}$. The atmosphere ($1 \text{ atm} = 101.325 \text{ kPa}$) and the bar, if used, should be in parenthesis, after the unit has been first expressed in Pascals, e.g. $1.23 \times 10^6 \text{ Pa}$ (12.3 atm). Manometric pressures in meters or millimeters are acceptable if one is reporting raw experimental results; otherwise they should be converted to Pa.

Velocity

Velocity is measured in m s^{-1} . Popular units such as km h^{-1} may be in parentheses afterward.

Volume

Volumes are measured in m^3 or liters ($1 \text{ liter} = 10^{-3} \text{ m}^3$). Abbreviations should not be used for the liter.

Flow

In S.I. units, flow should be expressed in kg s^{-1} , $\text{m}^3 \text{s}^{-1}$, liter s^{-1} . If non-standard units such as liter min^{-1} or kg h^{-1} must be used, they should be in parentheses afterward.

Temperature

The S.I. unit is the degree Kelvin (K). However, it is also permissible to express temperatures in the degree Celsius ($^{\circ}\text{C}$). Temperature differences are best expressed in Kelvin (K).

When compound units involving temperature are used, they should be expressed in terms of Kelvin, e.g. specific heat $\text{J kg}^{-1} \text{K}^{-1}$.

2. NOMENCLATURE AND SYMBOLS

Tables 1–5 list recommended symbols for physical quantities. Obviously, historical usage is of considerable importance in the choice of names and symbols and attempts have been made to reflect this fact in the tables. But conflicts do arise between lists that are derived from different disciplines. Generally, a firm recommendation has been made for each quantity, except for radiation where two options are given in Table 5.

In the recommendations for *material properties* (see Table 1), the emission, absorption, reflection, and transmission of radiation by

Table 1. Recommended symbols for materials properties

Quantity	Symbol	Unit
Specific heat	c	$\text{J kg}^{-1} \text{K}^{-1}$
Thermal conductivity	k	$\text{W m}^{-1} \text{K}^{-1}$
Extinction coefficient [†]	K	m^{-1}
Index of refraction	n	
Absorptance	α	
Thermal diffusivity	α	$\text{m}^2 \text{s}^{-1}$
Specific heat ratio	γ	
Emissance	ϵ	
Reflectance	ρ	
Density	ρ	kg m^{-3}
Transmittance	τ	

[†]In meteorology the *extinction coefficient* is the product of K and the path length, and is thus dimensionless.

Table 2. Recommended symbols and sign convention for sun and related angles

Quantity	Symbol	Range and sign convention
Altitude	α	0 to $+90^\circ$
Surface tilt	β	0 to $\pm 90^\circ$; towards the equator is +ive
Azimuth (of surface)	γ	0 to 360° ; clockwise from North is +ive
Declination	δ	0 to $\pm 23.45^\circ$
Incidence (on surface)	θ, i	0 to $+90^\circ$
Zenith angle	θ_z	0 to $+90^\circ$
Latitude	ϕ	0 to $\pm 90^\circ$; North is +ive
Hour angle	ω	-180° to $+180^\circ$; solar noon is 0, afternoon is +ive
Reflection (from surface)	r	0 to $+90^\circ$

Table 3. Recommended symbols for miscellaneous quantities

Quantity	Symbol	Unit
Area	A	m^2
Heat transfer coefficient	h	$\text{W m}^{-2} \text{K}^{-1}$
System mass	m	kg
Air mass (or air mass factor)	M	
Mass flow rate	$\dot{m}$	kg s^{-1}
Heat	Q	J
Heat flow rate	$\dot{Q}$	W
Heat flux	q	W m^{-2}
Temperature	T	K
Overall heat transfer coefficient	U	$\text{W m}^{-2} \text{K}^{-1}$
Efficiency	η	
Wavelength	λ	m
Frequency	ν	s^{-1}
Stefan-Boltzmann constant	σ	$\text{W m}^{-2} \text{K}^{-4}$
Time	t, τ, θ	s

materials have been described in terms of quantities with suffixes “ance” rather than “ivity,” which is also sometimes used, depending on the discipline. It is recommended that the suffix “ance” be used for the following four quantities:

Table 4. Recommended subscripts

Quantity	Symbol
Ambient	a
Black-body	b
Beam (direct)	b
Diffuse (scattered)	d
Horizontal	h
Incident	i
Normal	n
Outside atmosphere	o
Reflected	r
Solar	s
Solar constant	sc
Sunrise (sunset)	sr, (ss)
Total or global	t
Thermal	t, th
Useful	u
Spectral	λ

Table 5. Recommended symbols for radiation quantities

Preferred name	Symbol	Unit
(a) Nonsolar radiation		
Radiant energy	Q	J
Radiant flux	ϕ	W
Radiant flux density	ϕ	W m^{-2}
Irradiance	E, H	W m^{-2}
Radiosity or Radiant exitance	M, J	W m^{-2}
Radiant emissive power (radiant self-exitance)	M_s, E	W m^{-2}
Radiant intensity (radiance)	L	$\text{W m}^{-2} \text{sr}^{-1}$
Irradiation or radiant exposure	H	J m^{-2}
(b) Solar radiation		
Global irradiance or solar flux density	G	W m^{-2}
Beam irradiance	G_b	W m^{-2}
Diffuse irradiance	G_d	W m^{-2}
Global irradiation	H	J m^{-2}
Beam irradiation	H_b	J m^{-2}
Diffuse irradiation	H_d	J m^{-2}
(c) Atmospheric radiation		
Irradiation	$\phi \downarrow$	W m^{-2}
Radiosity	$\phi \uparrow$	W m^{-2}
Exchange	ϕ_N	W m^{-2}

$$\text{emittance } \epsilon = \frac{E}{E_b} \left(\text{or } \frac{M_s}{M_{sb}} \right)$$

$$\text{absorptance } \alpha = \frac{\phi}{\phi_i}$$

$$\text{reflectance } \rho = \frac{\phi}{\phi_i}$$

$$\text{transmittance } \tau = \frac{\phi}{\phi_i}$$

where E and ϕ is the radiant flux density that is involved in the particular process. The double use of α for both absorptance and thermal diffusivity is usual, as is the double use of ρ for both reflectance and density. Neither double use should give much concern in practice.